

BOLIN ZHENG

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EDUCATION

The University of Melbourne

Bachelor of Science in Data Science

March 2020 – January 2023

Melbourne, Australia

Monash University

Master of Information Technology

August 2023 – December 2024

Melbourne, Australia

TECHNICAL SKILLS

Programming Languages: Python, Java, R, SQL, C, C#, MATLAB

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Linux, CI/CD

Front-end: React, JavaScript, HTML5, CSS3, ASP.NET Core MVC

Back-end & Databases: Django, PostgreSQL, MySQL, DynamoDB

EXPERIENCE

Universe Action Limited

Data Engineer (AI & Infrastructure)

May 2025 – Present

Kowloon, Hong Kong

- **Enterprise GenAI Solutions:** Deployed **Enterprise LLM Solutions** focusing on **Natural Language Understanding (NLU)** for massive unstructured datasets. Leveraged **Prompt Engineering** and **RLHF** strategies to align model outputs with strict business rules, achieving a **90% accuracy rate**.
- **High-Performance ETL Pipeline:** Architected a scalable data processing pipeline within a **containerized Linux environment** using shared storage volumes. Handled **2TB+ of unstructured data** with **Python multi-threading** and **Automated Data Quality Checks**, reducing manual intervention by **90%** and ingestion errors significantly.
- **Model Governance & Compliance:** Implemented **Governance & Safety Guardrails** within the generative AI workflow. Designed logic to ensure compliance with project requirements, significantly **reducing model hallucinations** and safety risks.
- **Infrastructure Ownership (DevOps):** Took **Full Ownership** of the model lifecycle. Optimized model inference latency and VRAM usage, enabling high-throughput processing on constrained infrastructure, maximizing VRAM efficiency and ensuring system stability under high-concurrency scenarios.
- **Full-Stack Internal Tools:** Developed a proprietary **Data Management Dashboard** using **React and Django**, enabling the operations team to visualize data streams and manage annotation workflows efficiently.

Cura Education

AI Engineer Intern

November 2024 – April 2025

Melbourne, Australia

- **Automated Content Generation Pipeline:** Engineered an end-to-end Python pipeline to automate the workflow from data extraction to final content generation, increasing content production efficiency through **scalable AI-driven mechanisms**.
- **Prompt Engineering & Optimization:** Optimized OpenAI API prompts for multimodal tasks (Text/Image), improving model evaluation accuracy from 80% to **95%** through iterative testing and performance metrics analysis.
- **Database & SQL Management:** Utilized **PostgreSQL** to manage large-scale educational datasets. Wrote complex SQL queries to extract and structure data for model training.
- **Unstructured Data Parsing:** Developed a parsing engine to convert unstructured **HTML** raw data into structured **Markdown** format, enhancing data readability and compatibility for LLM ingestion.

International Digital Economy Academy (IDEA)

Data Operations Intern

March 2023 – June 2023

Shenzhen, China

- **Web Scraping for Alternative Data:** Developed **Python-based web crawlers** to acquire external datasets, enriching the training corpus and improving the model's domain adaptability.
- **Data Annotation Schema Design:** Collaborated on the architectural design of data annotation schemas and pre-annotation logic, defining the standards for high-quality datasets.
- **Model Performance Analysis:** Conducted qualitative analysis on model outputs (Chinese LLM), identifying failure cases and providing feedback loops to the algorithm team for model iteration.
- **Data Quality Assurance (QA):** Established data quality protocols to evaluate and correct model errors, ensuring the dataset aligned with specific business application scenarios.

PROJECTS

Melbourne Urban Green Space Finder | *React, Django, AWS (EC2), PostgreSQL, Google Maps API*

August 2024

- **Full-Stack Geo-Spatial Platform:** Architected a full-stack web application for analyzing urban data. Integrated **Google Maps API** and **Mapbox** to visualize real-time crowd density and geographical data, enabling interactive data exploration.
- **Quantitative Algorithm Implementation:** Designed and implemented backend algorithms to calculate carbon emission metrics for different transport modes. Processed real-time congestion data to generate dynamic route recommendations, reducing user carbon footprint by 15%.
- **Robust Backend API Services:** Developed high-performance **RESTful APIs** using **Django REST Framework**. Integrated multiple third-party data sources to fetch and aggregate real-time traffic and environmental data.
- **Production-Grade Deployment:** Configured and deployed the application on **AWS EC2** using **NGINX** as a reverse proxy and **Gunicorn** as the application server, ensuring high availability and security for concurrent user access.